Follow-Ahead Human-Robot Navigation via Monte Carlo Tree Search and Deep Reinforcement Learning

Archita Srivastava, Ankush Singh, Gemmin Sugiura, Jonathan Ung

***Abstract*—We replicate a methodology for robotic follow-ahead applications that addresses obstacle and occlusion avoidance. A high-level decision-making algorithm generates short-term navigational goals by integrating Monte Carlo Tree Search (MCTS) with Deep Reinforcement Learning (DRL). Experiments demonstrate that MCTS-DRL outperforms standalone MCTS and DRL baselines across obstacle-free and obstacle-present environments.**

***Index Terms*—human-robot interaction, follow-ahead, MCTS, deep reinforcement learning, obstacle avoidance**

I. Introduction

Human-following robots (HFR) maintain a consistent distance and orientation relative to a target person. Applications include autonomous luggage carts, search and rescue, activity filming, and patient monitoring. HFRs are broadly categorized as follow-behind, side-by-side, or follow-ahead. While follow-behind and side-by-side have received substantial attention [1]–[3], follow-ahead is more challenging because the robot must predict human intent and navigate proactively.

Prior follow-ahead work includes Moustris et al. [4], who used intention recognition for front-following. Nikdel et al. [5] combined DRL-based trajectory estimation with classical planners, referred to as LBGP. Mahdavian et al. [6] trained a non-autoregressive transformer for intention prediction. Obstacle and occlusion avoidance in follow-behind or side-by-side scenarios has been explored using lidar and cameras with PD and MPC controllers [7]–[12]; however, no existing follow-ahead method handles this problem.

We replicate the MCTS-DRL approach of [13], the first follow-ahead method to address obstacle and occlusion avoidance. The main contributions are: (1) a follow-ahead method that avoids collisions and occlusions in complex environments; (2) a high-level decision-making algorithm that adapts MCTS to a continuous path-planning setting; and (3) integration of DRL into MCTS to produce consistent navigational goals.

II. Methods

A. State, Observation, and Action Space

The system state is state $= (x_r, y_r, \theta_r, x_h, y_h, \theta_h)$, where x, y, θ denote 2D pose and subscripts r, h refer to robot and human. Each MCTS node represents one such state. The DRL model uses the relative observation $o = (x_h - x_r,\ y_h - y_r,\ \theta_h - \theta_r)$.

Three discrete actions are defined per time step ($\delta t = 0.5$ s): go straight, turn left $45°$, or turn right $45°$. The robot's next pose under action $\psi \in \{-45°, 0°, 45°\}$ is

$$\begin{bmatrix} x_r' \\ y_r' \\ \theta_r' \end{bmatrix} = \begin{bmatrix} x_r \\ y_r \\ \theta_r \end{bmatrix} + \begin{bmatrix} d\cos(\theta_r + \psi) \\ d\sin(\theta_r + \psi) \\ \psi \end{bmatrix} \tag{1}$$

assuming constant velocity $V = 0.6$ m/s, giving $d = 0.3$ m per step.

B. Reward Function

The reward encourages the robot to remain in front of the person within a distance range of $[1, 2]$ m:

$$r = \max(r_d + r_\alpha,\ -1) \tag{2}$$

where $r_\alpha = (45 - \alpha)/45$ penalizes angular deviation, and r_d penalizes distance error: $r_d = -(1 - d_h)$ if $0.5 < d_h < 1$; 0 if $1 \le d_h \le 2$; $-0.25(d_h - 1)$ if $2 < d_h < 5$; and -1 otherwise. Here α is the angle between the person-robot vector and the person's heading, and d_h is the human-robot distance.

C. DRL Training

A Double DQN (DDQN) [14] is trained offline in an obstacle-free simulation. A dataset of ten million tuples (o, a, R, o') is generated with randomized human trajectories and initial robot poses. The return is $R = \sum_{i=0}^{n} \gamma^i r_i$ with $\gamma = 0.99$. The trained model outputs $R_i = Q(o, a_i)$ for each action, providing value estimates used during MCTS tree expansion.

D. MCTS-DRL Integration

At each planning step, MCTS expands a search tree over a $T = 3$ s horizon in a receding-horizon fashion. The root node encodes the current robot and human pose. Child nodes are generated via (1) and pruned if they collide with obstacles (via the occupancy map from SLAM-gmapping [15]) or cause occlusion. Non-pruned nodes are assigned value $R_i = Q(o, a_i)$ from the DRL model; pruned nodes receive -1.

Node selection follows the Upper Confidence Bound:

$$\text{UCB} = \frac{w}{n^c} + c\sqrt{\frac{\log n^p}{n^c}} \tag{3}$$

where w is the node value (equivalent to Q), n^c and n^p are visit counts of the child and parent nodes, and $c = 1.4$ balances

Algorithm 1: MCTS-DRL Approach

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Data: robot pose = (x_r, y_r, θ_r)^{t_0}
Data: (x_h, y_h, θ_h)^{t_0}, ..., (x_h, y_h, θ_h)^{t_3}
Data: occupancy map
Result: navigational goal point
parent node = (x_r^{t_0}, y_r^{t_0}, θ_r^{t_0}, x_h^{t_0}, y_h^{t_0}, θ_h^{t_0});
for num of expansion do
    while parent node is not fully expanded do
        child ← simulate using an action;
        if no collision then
            if no occlusion then
                R ← Q(o, a_i);
                child node state ← simulate using the
                  action;
            else
                value ← −1;
            parent value ← parent value + value;
        else
            delete the child node;
    parent node ← leaf node with highest UCB;
goal point ← leaf node with highest UCB (c = 0);
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exploration and exploitation. After tree expansion, the leaf node with the highest UCB (evaluated at $c = 0$) is selected as the robot's navigational goal. The ROS navigation stack with TEB planner [16] handles low-level execution.

III. Experiments

Three experiments evaluate the method in simulation and on a real robot.

A. MCTS-DRL vs. MCTS and DRL

Fig. 1. Placeholder: trajectory comparison of DRL, MCTS, and MCTS-DRL for circular and S-shaped paths.

In simulation, MCTS-DRL is compared to standalone MCTS and DRL on circular and S-shaped human trajectories. DRL fails to avoid obstacles since it was trained in obstacle-free settings. MCTS produces inconsistent trajectories near corners due to random action sampling. MCTS-DRL maintains a stable trajectory ahead of the human while avoiding obstacles. Table I summarizes cumulative rewards over 20 trials.

B. Follow-Ahead in an Obstacle-Free Environment

MCTS-DRL is compared to LBGP [5] across straight, S-shaped, and U-shaped trajectories in a real-world obstacle-free environment. Human-robot relative distance and mean orientation angle α are measured using a motion capture system. Table II shows that MCTS-DRL maintains distance within $[1, 2]$ m across all trajectories, achieving comparable performance to LBGP.

TABLE I
Cumulative rewards for DRL, MCTS, and MCTS-DRL.

Trajectory	DRL	MCTS	MCTS-DRL
Circle	X	$X \pm X$	**X**
S-shaped	X	$X \pm X$	**X**

TABLE II
Obstacle-free comparison: mean distance and orientation vs. LBGP.

Trajectory	Method	Distance (m)	α (deg)
Straight	MCTS-DRL	$X \pm X$	$X \pm X$
	LBGP	$X \pm X$	$X \pm X$
S-shaped	MCTS-DRL	$X \pm X$	$X \pm X$
	LBGP	$X \pm X$	$X \pm X$
U-shaped	MCTS-DRL	$X \pm X$	$X \pm X$
	LBGP	$X \pm X$	$X \pm X$

C. Obstacle and Occlusion Avoidance

Fig. 2. Placeholder: robot trajectories with and without obstacles for straight, U-shaped, and S-shaped paths.

Since no prior follow-ahead method handles obstacles, this experiment evaluates MCTS-DRL against itself with and without obstacles. Across all three trajectories, the robot successfully follows ahead in the obstacle-free case. When obstacles are introduced, the robot alters its path to avoid occlusion rather than rotating around the obstacle. In an L-shaped corridor, the robot temporarily moves beside the human to avoid a corner collision, then resumes follow-ahead navigation.

IV. Conclusion

We replicated an MCTS-DRL framework for follow-ahead human-robot navigation with obstacle and occlusion avoidance. By combining DRL-based value estimation with MCTS tree search, the method generates consistent navigational goals in environments with arbitrary obstacles. Experiments confirm that MCTS-DRL outperforms standalone MCTS and DRL, and performs comparably to LBGP in obstacle-free settings while additionally handling real-world obstacles.

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